

MACON COUNTY AP, NC, USA

WMO: 720259

Lat: 35.223N

Lon: 83.419W

Elev: 2020

StdP: 13.65

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 63844

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 14.0	(c) 18.2	(d) -3.7	(e) 4.9	(f) 31.2	(g) 2.3	(h) 6.7	(i) 31.9	(j) 17.2	(k) 38.3	(l) 15.9	(m) 37.9	(n) 2.1	(o) 350	(p) 0.374

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.6	(c) 89.5	(d) 69.4	(e) 87.5	(f) 68.9	(g) 85.5	(h) 68.9	(i) 73.5	(j) 82.1	(k) 72.5	(l) 80.9	(m) 71.4	(n) 79.7	(o) 4.5	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.4	125.1	76.6	70.1	119.6	75.6	69.1	115.3	74.8	38.6	82.7	37.5	80.4	36.5	80.4	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	11.6	10.0		7.7	92.5	5.4	3.4	3.8	94.9	0.7	96.9	-2.4	98.8	-6.3	101.2
				6.8	75.2	5.2	1.8	3.0	76.5	-0.1	77.6	-3.0	78.6	-6.8	79.9
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.9	38.2	41.5	48.2	56.3	63.9	71.5	74.1	73.2	68.4	57.7	47.1	41.7
	DBStd	14.55	9.62	8.95	8.25	6.91	6.19	4.04	3.64	3.97	4.88	7.94	7.89	8.65
	HDD50	1254	381	262	134	23	2	0	0	0	22	145	285	
	HDD65	3912	830	658	521	265	93	5	1	26	250	539	723	
	CDD50	3766	16	24	77	212	434	645	747	718	551	259	56	27
	CDD65	952	0	0	0	4	60	200	283	254	127	23	1	0
	CDH74	7526	0	1	35	236	720	1672	2007	1748	915	183	9	0
	CDH80	2229	0	0	2	27	152	550	685	544	243	26	0	0
	WSAvg	2.2	2.9	3.3	3.3	3.3	2.2	1.6	1.2	1.1	1.1	1.8	2.2	2.3
	PrecAvg	70.1	7.0	6.5	7.0	5.3	4.6	6.2	5.9	5.6	5.7	4.2	5.3	6.9
(15) Precipitation	PrecMax	86.3	13.5	12.2	12.0	9.3	10.1	12.1	10.1	11.1	18.4	10.4	12.0	11.4
	PrecMin	53.1	2.8	3.3	2.1	2.2	0.9	1.6	2.8	1.1	0.3	0.2	2.0	4.1
	PrecStd	11.0	2.6	2.2	2.5	1.9	2.2	2.5	2.1	2.7	4.4	2.5	2.3	1.9
	PrecStd	11.0	2.6	2.2	2.5	1.9	2.2	2.5	2.1	2.7	4.4	2.5	2.3	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	66.3	71.7	78.4	82.8	87.7	91.2	90.9	90.9	88.3	83.1	75.0	69.1	
	0.4% MCWB	54.1	59.2	56.9	59.1	63.7	67.2	70.4	70.7	67.8	66.6	57.7	57.2	
	2% DB	63.0	66.1	73.2	80.0	84.0	88.4	88.4	88.3	85.8	78.8	70.2	63.9	
	2% MCWB	53.5	55.1	55.1	58.2	64.6	67.7	70.4	70.3	67.4	62.5	55.6	56.1	
	5% DB	59.0	62.7	69.7	76.8	81.3	86.0	86.4	85.9	82.4	75.2	66.2	60.8	
	5% MCWB	52.5	51.6	55.0	57.8	63.1	68.0	70.0	70.2	66.3	60.7	54.9	54.0	
	10% DB	54.5	58.3	64.8	73.0	78.6	82.6	84.1	82.7	79.9	72.1	62.8	56.8	
	10% MCWB	49.2	48.9	51.8	56.6	63.0	66.8	69.7	68.8	65.7	60.1	52.9	51.7	
	0.4% WB	58.7	61.6	61.8	66.3	71.9	72.8	75.2	75.2	72.1	70.3	64.2	62.2	
	0.4% MCDB	61.8	67.2	71.9	73.9	81.3	81.8	84.4	85.2	81.0	77.4	69.3	65.7	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	56.1	58.4	59.1	63.4	69.0	71.4	73.7	73.6	70.1	67.5	61.6	59.2	
	2% MCDB	61.0	64.4	68.8	71.5	77.0	81.4	82.3	82.3	78.7	73.8	66.7	62.7	
	5% WB	53.6	54.8	57.0	61.5	67.2	70.4	72.5	72.6	69.0	65.5	58.2	55.6	
	5% MCDB	57.2	59.5	65.6	71.3	75.4	80.2	81.0	80.7	77.2	71.8	62.7	59.0	
	10% WB	50.3	51.0	54.6	59.6	65.7	69.2	71.5	71.4	67.8	62.7	55.1	52.7	
	10% MCDB	53.6	55.8	61.7	68.6	73.6	78.7	79.8	78.9	75.6	68.6	59.7	55.8	
(35) Mean Daily Temperature Range	MDBR	22.5	23.4	25.4	27.6	25.0	22.8	20.6	20.8	22.3	25.5	27.2	22.2	
	5% DB	26.3	29.7	32.4	35.1	30.3	26.9	23.4	23.0	25.9	28.0	33.0	24.9	
	5% WB	17.5	17.8	16.5	16.0	12.9	10.0	8.6	8.5	10.5	13.6	18.6	16.5	
	MCDBR	19.6	21.4	26.6	25.8	22.5	22.1	20.0	18.9	21.8	21.4	20.4	17.8	
	MCWBR	15.5	15.7	15.8	14.1	11.2	9.2	8.2	7.8	9.7	11.4	13.9	13.9	
	MCWBR	15.5	15.7	15.8	14.1	11.2	9.2	8.2	7.8	9.7	11.4	13.9	13.9	
(40) Clear-Sky Solar Irradiance	taub	0.300	0.310	0.341	0.377	0.433	0.459	0.497	0.500	0.395	0.362	0.322	0.299	
	taud	2.533	2.501	2.437	2.337	2.201	2.155	2.075	2.050	2.329	2.415	2.492	2.561	
	Ebn at Noon	289	298	295	288	272	264	252	249	273	274	277	282	
	Edh at Noon	25	29	34	40	46	48	52	52	37	32	26	23	
(44) All-Sky Solar Radiation	RadAvg	796	978	1330	1665	1834	1926	1830	1703	1478	1220	904	686	
	RadStd	56	114	98	120	126	140	132	90	132	132	91	66	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (36 neighbors)	+0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+261	+173

Nomenclature: See separate page